Customer Shopping Behavior Analysis

Python → SQL → Power BI: An End-to-End Analytics Case Study

Prepared by: **Tahseenah Hamza** | Tools: Python (pandas), PostgreSQL, Power BI Desktop | Dataset: 3,900 customer transaction records

Business Problem

> **Overarching question:** "How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

Shifting Patterns

Management observed changes in purchasing behavior across demographics, product categories, and sales channels.

Key Unknowns

Which factors — discounts, reviews, seasons, or payment preferences — actually drive consumer decisions and repeat purchases?

Why It Matters

Answering this unlocks smarter targeting, better retention, and higher long-term customer value.

Dataset

3,900 customer transaction records across categories, demographics, and purchase behavior.

Project Workflow

A three-stage end-to-end analytics pipeline — each tool chosen for what it does best.

Python Prep

SQL Testing

Power BI

Each stage hands off a refined artifact to the next: Python produces a clean dataset, SQL produces validated insights, and Power BI communicates those findings to stakeholders.

Data Cleaning Highlights

Five deliberate decisions in Python — each tied to how the data would be used downstream in SQL and Power BI.

Category-Level Imputation

Missing review ratings filled with per-category median — preserving the fact that Outerwear and Accessories ratings differ.

Column Standardization

All headers converted to `snake_case` — making every SQL query and Power BI field reference clean.

Age Group Engineering

Continuous age binned into 4 quartile-balanced segments (Young Adult → Senior).

Frequency Encoding

Text labels ("Weekly", "Monthly") mapped to numeric days — converting unusable categorical into quantitative feature.

Redundancy Removal

`promo_code_used` confirmed identical to `discount_applied` across all 3,900 rows, then dropped — verified before removed.

SQL Analysis Approach

10 business hypotheses tested in PostgreSQL — each framed as a question a retail stakeholder would actually ask, not a generic aggregate.

Gender Revenue Gap = Headcount, Not Behavior

Male: **$157,890** vs. Female: **$75,191** — but avg spend is nearly identical ($59.54 vs. $60.25). The gap is a volume story.

Discount Users Aren't Bargain Hunters

839 customers used a discount and still spent at or above the $59.76 average — roughly half of all discount users.

Subscription ≠ Higher Spend

Subscribers averaged **$59.49** vs. $59.87 for non-subscribers — essentially flat. Reframes the subscription program's value proposition.

Most Loyal Customers Aren't Subscribed

Among repeat buyers (5+ purchases): **2,518 non-subscribers** vs. only 958 subscribers — the single most actionable finding.

Power BI Dashboard

Design Principles

- **Variance-based chart selection:** Revenue by Category (CV 66%) kept; Age Group revenue (CV 5.2%) cut as too flat to be actionable.
- **Bar charts** for comparing absolute magnitudes across 4+ categories; **donut charts** for part-to-whole composition with 2–4 segments.

Dashboard Components

- **KPI row:** Total Revenue, Total Customers, Avg Purchase Amount, Avg Rating — business health at a glance.
- **Interactive filters:** Subscription Status, Gender, Category, Size — lets any viewer re-slice the story.

Key Insights

1

Loyal Customers Are Under-Subscribed

72% of repeat buyers (5+ purchases) are not enrolled in the subscription program — the company's most valuable customers are its least captured.

2

Subscription ≠ Higher Spend

Avg spend is nearly identical: subscribers **$59.49** vs. non-subscribers **$59.87** — the program's value proposition needs re-evaluation.

3

Discounting Is Uniform, Not Strategic

Discount rates cluster at **45–50%** across top products with no standout items — not a differentiated lever for driving above-average spend.

4

Category > Demographics as Segmentation Axis

Clothing vs. Outerwear shows a **5.6× revenue spread**; age group shows only 10% spread — category is the stronger lens.

5

Loyalty-Heavy Base

79.9% of customers are "Loyal"; only **2.1%** are "New" — acquisition is not the constraint, retention conversion is.

Recommendation

Target the ~2,500 Non-Subscribed Repeat Buyers

These customers already demonstrate loyalty through repeat behavior — making them a lower-risk, higher-conversion audience than acquiring new subscribers from scratch.

01

Who

Customers with 5+ previous purchases not currently enrolled in the subscription program.

02

What to Fix First

Re-evaluate subscription benefits — flat spend difference signals enrollment alone won't move the needle without a stronger value proposition.

03

Next Step

A/B test a targeted incentive offer to this segment and measure conversion rate and post-enrollment spend lift.

Thank You

Questions, feedback, or collaboration — reach out via the links below.

Prepared by	GitHub	Tools Used
Tahseenah Hamza	Full project code, SQL queries, and cleaned dataset available in the project repository.	Python (pandas, Jupyter) · PostgreSQL · Power BI Desktop